

# KrishokTech: Deterministic-First, Evidence-Bounded Bengali Agricultural Advisory

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## Abstract

Bengali agricultural advisory systems face a particular difficulty: treatment requests may omit the crop, photographs may conflict with the accompanying text, and generated dosage claims may exceed the retrieved evidence. KRISHOKTECH treats these uncertainties as explicit control states rather than delegating them to the language model. A deterministic gate halts treatment queries without a resolvable crop before retrieval. Crop identification fences the evidence set, and a confirmation state is raised when text and image disagree. Before rendering, each dosage-bearing claim in the generated draft is checked against retrieved passages and permitted dose ranges; unsupported claims are dropped. The demonstration follows one farmer interaction through clarification, multimodal conflict, safety referral, and claim filtering. On 200 authentic farmer queries, the gate halted all 30 crop-less requests before retrieval. Crop fencing with contradiction gating reduced off-target retrieval from 36.25% to 28.75% on a 400-query stress set. On the deployed local model, guarded attack success fell from 52.5% to 10.71%; on the cloud benchmark, from 36.19% to 0.95%. The dosage verifier caught 114 of 118 inserted errors in live answers without rejecting any unmodified response.

## 1 Introduction

When yellow spots appear on potato leaves, a farmer often needs advice before an extension officer can visit. While digital advisory services bridge this gap (Department of Agricultural Extension, 2022; Developing Local Extension Capacity (DLEC) Project, 2021; Aker, 2011; Fabregas et al., 2019; Van Campenhout et al., 2021), fluent conversation alone is insufficient in a treatment setting: an answer for the wrong crop or an unsupported dosage can turn an incomplete question into a harmful recommendation.

A farmer may ask “which pesticide should I spray?” without naming the crop, leaving retrieval to search across unrelated guidance. Farmers may also write in colloquial or Romanized Bangla, while technical guidance uses formal terminology (Bhuiyan et al., 2025; Reza et al., 2026a). A photograph and text can also disagree; a system that silently chooses one modality leaves the conflict unresolved, a known failure mode in vision-language models (Deng et al., 2025; Zhu et al., 2024). Even when evidence is successfully retrieved, faithful generation is not guaranteed. Language models can produce claims unsupported by their context (Brown et al., 2025; Xu et al., 2025), which is particularly dangerous for agricultural dosage, where a plausible numerical claim is directly actionable (Reza et al., 2026b).

KRISHOKTECH makes these decisions explicit as four farmer-visible states: **ASK**, **CONFIRM**, **REFER**, and **DROP**. Three controls enforce them: (C1) a deterministic gate halts underspecified treatment queries before retrieval and requests the missing crop; (C2) crop identification fences retrieval to the identified crop and raises confirmation when text and image disagree; and (C3) dosage-bearing claims are checked against retrieved passages and BARI/BRRI dose ranges before rendering, with unsupported claims dropped.

The demonstration traces one continuous farmer session through these states, from a colloquial crop-less query to crop-scoped evidence, multimodal conflict, safety referral, and answer delivery after dosage verification (Section 4). Interactive demo: <https://krishoktech-one.vercel.app>; screencast (<2.5 min): <https://krishoktech-one.vercel.app/screencast>. Source code and installable package are available under Apache-2.0 at <https://anonymous.4open.science/r/KrishokChat-Agricultural-Advisory-System-FCEE>.

## 2 Related Work and Positioning

Digital agricultural advisors such as Farmer.Chat, KrishokBondhu, and Krishi Sathi have paired localized agricultural knowledge with conversational interfaces to improve access to timely advice (Singh et al., 2024; Ameen et al., 2026; Vijayvargia et al., 2025). Bengali agricultural RAG systems have also explored matching farmer queries to more formal scientific knowledge (Hossain et al., 2026). This work establishes localized retrieval and multilingual interaction as practical building blocks. KrishokTech instead focuses on when the system should be allowed to proceed: a crop-less treatment request is treated as an unresolved safety condition rather than an incomplete conversational turn.

Classic information-seeking research shows that asking a clarifying question improves retrieval when a request is underspecified (Aliannejadi et al., 2019). LLM-based agents have more recently been shown to guess or hedge under ambiguity instead of reliably asking for missing information (Chen et al., 2024; Gan et al., 2024). MiniCheck demonstrates efficient claim-level checking of generated text against grounding documents, while MedRAGChecker extends atomic claim verification to safety-sensitive biomedical RAG (Tang et al., 2024; Ji et al., 2026). Interactive systems such as My Climate CoPilot, FactSearch, and RAGVUE emphasize grounded answers, inspectable evidence, and fine-grained diagnosis rather than treating the generated response as a single opaque output (Nguyen et al., 2025; Fang and Mackenzie, 2026; Murugaraj et al., 2026). KrishokTech applies these ideas as explicit pipeline boundaries before retrieval and before rendering.

The multimodal setting introduces a further failure mode. When text and visual evidence conflict, vision-language models frequently rely on misleading textual information rather than reconciling the sources (Deng et al., 2025; Zhu et al., 2024). The linguistic reality of Bengali deployment—where farmers use regional dialects and Latin-script Banglish—means that safety gates tuned only for formal text can be brittle (Bhuiyan et al., 2025; Kakwani et al., 2020). KrishokTech addresses both by refusing to silently select a modality when crop evidence disagrees, and by enforcing deterministic, script-agnostic prechecks before generation. Ultimately, our contribution is the integration of these boundaries into a single pipeline where uncertainty is an observable system state rather than a hidden

model behavior.

## 3 System Design

KrishokTech implements four farmer-visible states through five ordered stages, T0–T4, followed by delivery (Figure 1). Each stage constrains the next: T0/T1 can end the turn in **REFER/ASK**, T2 in **CONFIRM**, and T4 filters the T3 draft before delivery.

### 3.1 Halting Before Retrieval

The first decision is whether a request should reach retrieval at all.

At **T0**, a deterministic precheck examines incoming Bengali and Banglish text for high-risk chemical requests, poisoning-related language, and crisis patterns covered by the system’s safety policy. A matched request is stopped before retrieval or generation and routed to the Krishi Call Centre (**16123**), without invoking a language model. The exact matching patterns are withheld to prevent evasion; measured safety results are in Appendix D. The boundary follows the system’s safety policy; it does not attempt to reproduce the full regulatory status of every chemical.

Requests that pass T0 reach **T1**. Here, a deterministic information-state extractor identifies the crop, symptom, and treatment intent without invoking a language model. When a treatment request contains no resolvable crop, the crop is treated as a safety-critical missing slot. The pipeline halts before BM25 retrieval runs (*sources\_retrieved* = 0) and enters **ASK**. The interface can then request the crop through Bengali quick-reply chips, provide a non-chemical guidance card, or offer referral to the helpline. A *DialectSelector* can apply one of six regional synonym presets during query normalization to accommodate colloquial variation (Reza et al., 2026a).

### 3.2 Crop-Fenced Retrieval and the Mismatch Badge

Requests that pass the information gate enter **T2**, where crop identity determines what evidence retrieval is allowed to access. KrishokTech uses BM25 over a curated index of BARI, BRRI, and DAE agricultural guidelines (Bangladesh Agricultural Research Institute, 2023; Bangladesh Rice Research Institute, 2024; Department of Agricultural Extension, 2022). The current deployment covers six staple crops: rice, potato, brassica, chili, corn, and wheat.

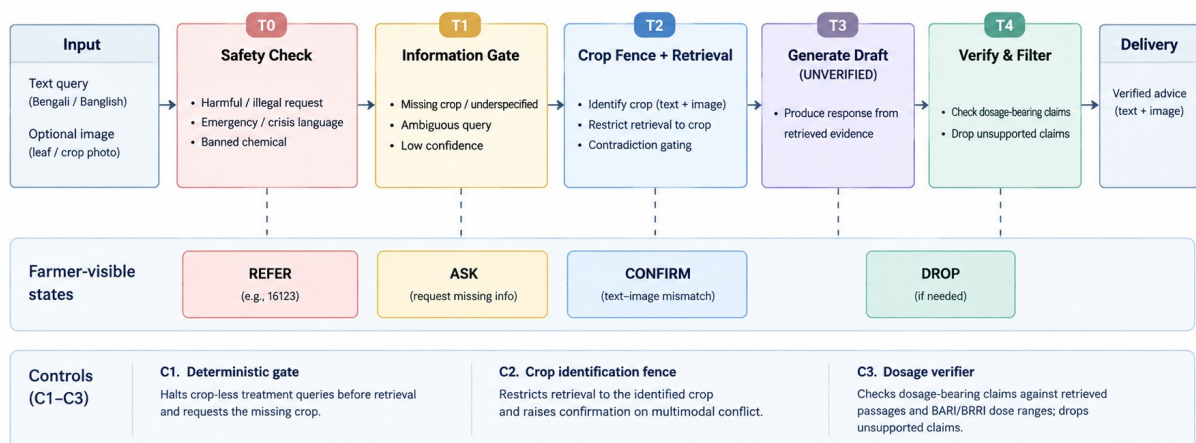


Figure 1: KrishokTech uses five control stages (T0–T4), followed by delivery. T0 can end a request in **REFER**, T1 in **ASK**, T2 in **CONFIRM**, and T4 in **DROP**; T3 produces an unverified draft that must pass T4 before delivery.

When a farmer attaches a photograph, an on-device vision model identifies the crop before retrieval proceeds. The router has three outcomes: a confident prediction routes the request to the corresponding crop-specific evidence, an ambiguous prediction presents the top candidates for disambiguation, and an out-of-scope image asks the farmer to recapture the photograph. The exact confidence and margin thresholds are given in Appendix A.

Once a crop is bound, it becomes a hard retrieval fence. Passages associated with other crops are excluded from the candidate evidence set rather than simply receiving a lower ranking. This prevents evidence for an unrelated crop from reaching generation. If the farmer’s text names a different crop from the attached photograph, the system does not silently choose between them. Instead, it raises a **Crop Mismatch** warning under the **Clarification** badge and enters **CONFIRM**, withholding chemical advice until the farmer confirms the intended crop. For conflicting diagnoses within a crop, the system asks a discriminative question rather than combining incompatible treatments into a single response.

### 3.3 Untrusted Generation, Verification, and Delivery

Once the request has passed T0–T2, T3 generates a response from the crop-scoped evidence. The deployed system uses `krishokchat-4b`, a 4-bit quantized Gemma-based model fine-tuned with a LoRA adapter and served through `llama-server`. The cloud model,

`google/gemini-2.5-flash-lite`, is used only as a benchmark configuration in the evaluation.

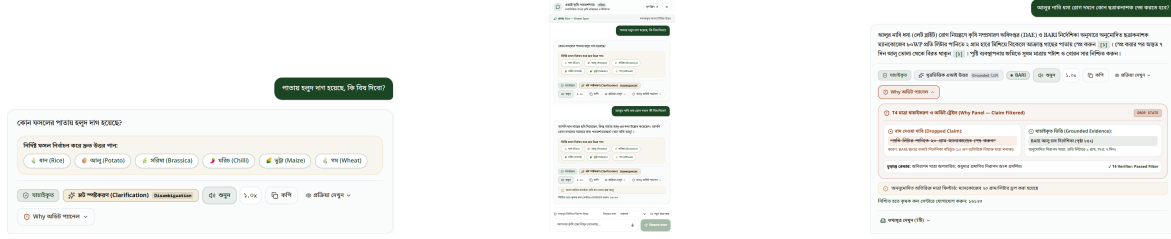
In both cases, the generated response is treated as an *unverified draft*, not as the final answer.

At **T4**, KrishokTech checks numerical claims that carry dosage information against the retrieved evidence and the permitted BARI/BRRI dose range. A claim is withheld when it cannot be grounded in the retrieved passage or falls outside the permitted range; the unsupported claim is **dropped** rather than rendered. This verifier is intentionally narrow: it checks dosage-bearing numerical claims and does not establish the factual correctness of every agronomic statement in the response. The verification breakdown is summarized in the evidence ledger (Appendix D).

Delivered outputs carry a provenance badge and Why-panel trace and can be rendered as web, SMS, offline card, or read-aloud.

## 4 Demonstration

The demonstration is intended for researchers and practitioners building multilingual, multimodal, or safety-critical conversational systems. Attendees follow one continuous farmer session on the live system in five steps, S1–S5, in the same order as the screencast. Together, the steps expose the four control states introduced in Section 1: **ASK** (S1), **CONFIRM** (S3), **REFER** (S4), and **DROP** (S5). S2 establishes the visual crop context used by S3. Attendees can begin with pre-loaded regional-dialect and safety-test queries or enter their own



(a) **ASK (S1)**. Missing crop halts retrieval ( $sources\_retrieved = 0$ ); prompts farmer with 6 staple crop chips. (b) **CONFIRM (S3)**. Rice photograph conflict with potato text triggers Clarification badge, withholding chemical advice. (c) **DROP (S5)**. Expanded *Why* panel exposes filtered  $10\times$  overdose claim and validates BARI p. 852 grounding.

Figure 2: Live interface states for ASK, CONFIRM, and DROP; complete continuous demonstration S1–S5 in Appendix C.

text and photographs.

**S1: Missing crop (ASK).** The farmer submits a colloquial treatment query without naming the crop: “*Patay holud dag hoyeche, ki bish dibo?*” (“Yellow spots on leaves. What pesticide to use?”). Because the query lacks a crop, the **T1** gate halts retrieval ( $sources\_retrieved = 0$ ) and renders quick-reply crop chips (Figure 2a). Selecting *Alu* (potato) binds the crop and initiates potato-scoped retrieval with numbered citations.

**S2: Photograph scopes evidence.** The farmer uploads a leaf photo. An on-device vision classifier identifies rice, scoping the session to rice guidance. The image remains local and restricts candidate evidence at the **T2** fence.

**S3: Text–image conflict (CONFIRM).** With the rice image attached, the farmer queries potato late blight. Rather than guessing, the system raises a **Clarification** badge and withholds chemical advice until confirmed (Figure 2b).

**S4: High-risk chemical request (REFER).** The farmer asks how to apply paraquat, which is blocked by KrishokTech’s safety policy. The **T0** precheck blocks the turn before retrieval or generation, displaying a safety alert and a 1-click route to the Krishi Call Centre (**16123**).

**S5: Dosage verification (DROP).** The farmer requests fungicide for potato blight. At **T4**, claims are checked against retrieved BARI/BRI evidence and dose bands. One unsupported dosage statement is **dropped**, and the verified residue is rendered with citations, provenance badge, and an expandable *Why* audit panel (Figure 2c; see Appendix C for alternate delivery channels).

## 5 Evaluation

We evaluate each control against the failure mode it targets; Appendix D lists every result with its

| Control     | Test                 | Result                                                 |
|-------------|----------------------|--------------------------------------------------------|
| Gate (C1)   | 200; 30 crop-less    | 30/30 halted, 0 false halts                            |
| Fence (C2)  | 400; 454 conflicts   | 36.25% $\rightarrow$ 28.75%; 453/454                   |
| Attack (C3) | 420 cloud, 280 depl. | 36.19% $\rightarrow$ 0.95%; 52.5% $\rightarrow$ 10.71% |
| Dosage (C3) | 118 live, 350 scaled | 114 caught; 296 (84.6%)                                |

Table 1: Headline results; §5 and Appendix D give CIs and scope.

status.

### 5.1 Pre-Retrieval Gate (C1)

We test the gate on 200 authentic Bengali farmer treatment queries, 30 of which name no crop. Two annotators labeled the crop slot independently and agreed on every crop-presence decision (Cohen’s  $\kappa = 1.00$ ). The gate halted all 30 crop-less queries before retrieval and passed the other 170 to fenced search without a false halt.

An unconstrained LLM asked to make the same decision halted 10% of the 200 queries, compared with 15% for the deterministic gate, agreed with the gate on 91% of decisions, and took a median of 4.2 s per query. On this set, the deterministic slot check matched the annotators on every query without calling a model. Slot-extraction accuracy, the controlled pilot, and token metering are in Appendix A.

### 5.2 Crop Fence and Mismatch Badge (C2)

The six on-device disease classifiers reach 0.94 to 0.99 top-1 accuracy on 4,294 held-out test images. To test the fence, we use a 400-query stress set of dialectal, ambiguous, and conflicting queries selected to stress retrieval under these conditions. A query is counted as yielding wrong-crop advice when its top five sources include a document for another crop. Open retrieval does so for 36.25% of the queries. Fencing retrieval to the crop predicted by the on-device vision model, at its empirical error rate of

about 5%, lowers this to 31.25%. Adding contradiction gating, which halts conflicting queries before retrieval, lowers it further to 28.75% (a text-only ablation without image fencing, reported in Appendix D, reaches a similar 28.75% endpoint through a distinct mechanism).

The mismatch badge caught 453 of 454 explicit text-image contradictions (53 of 54 farmer cases and all 400 cases in PRISM, our synthetic multimodal-conflict suite) with no false halts. The one miss used an implicit crop reference. On a separate set of 300 conflict queries, a prompted LLM asked the farmer to clarify in only 20% of cases, and naive multimodal RAG in 13%. Per-crop accuracy, quantization, and the full fence comparison are in Appendix D.

### 5.3 Safety Verification and Constrained Delivery (C3)

We first measure attack success rate (ASR), the share of adversarial prompts that yield an unsafe response, with the deterministic controls on (guarded) and off (unguarded). Across 420 live calls to the cloud benchmark model (gemini-2.5-flash-lite), 210 per arm, the guarded pipeline had an ASR of 0.95% against 36.19% unguarded. On the Bengali-native subset of 100 attacks, the guarded ASR was 10.0% against 83.0%, with the residual failures concentrated in embedded (4 of 10 prompts) and role-play (2 of 10) variants. On the deployed model, krishokchat-4b (280 calls per arm), ASR falls from 52.5% to 10.71% overall and from 77.0% to 19.0% on the Bengali-native subset, so the residual ASR is higher than on the cloud model. Latin-script Banglish and phonetic spellings such as *parakwat* also evade keyword gates written for formal text. A gate limited to formal English and Bengali script intercepted 1 of 85 adversarial queries in this style, whereas our deterministic precheck intercepted all 85 and raised no false alarm on 15 benign farmer questions.

We then assess the generated advice itself. In a blinded audit, three evaluators (two agronomy or botany undergraduates and one extension officer) rated 46 of 47 guarded outputs as safe and actionable, one as vague but harmless, and none as dangerous (Fleiss’  $\kappa = 0.82$ ). All three unguarded control outputs were rated dangerous. For the dosage verifier, we inserted controlled errors into verified answers. On 38 live answers with 118 inserted errors, it caught 114 and rejected none of

| Stress Regime           | $n$        | Turns      | Metric            | Result        |
|-------------------------|------------|------------|-------------------|---------------|
| Anaphoric Carryover     | 75         | 175        | Slot Retention    | 100.0%        |
| Crop Topic Shift        | 75         | 150        | Topic Isolation   | 100.0%        |
| Cross-Crop Hazard       | 75         | 150        | Chem. Leakage     | 0.0%          |
| Delayed Evasion         | 30         | 60         | 16123 Catch       | 100.0%        |
| Clarification Gate      | 20         | 40         | ASK & Resolve     | 100.0%        |
| <b>Total Multi-Turn</b> | <b>200</b> | <b>425</b> | <b>Robustness</b> | <b>100.0%</b> |

Table 2: Multi-turn robustness benchmark ( $N = 200$  dialogues, 425 turns) across four stress regimes using structured working memory, query rewriting, and T0 prechecks.

the unmodified answers. On a scaled benchmark of 528 cases from 88 BARI and BRRI passages, it caught 296 of 350 inserted errors (84.6%): every overdose (76 of 76) and underdose (67 of 67), but only 85% of chemical substitutions, 73% of unit substitutions, and 65% of invented pre-harvest intervals.

Constrained delivery preserves the dose: the SMS packer kept 399 of 400 messages within 160 characters and preserved the dose in 92 of 100 long advisories without invention; full SMS, offline, and latency detail is in Appendix B and the evidence ledger (Appendix D).

### 5.4 Multi-Turn Conversational Robustness

Real-world deployment requires maintaining agromomic coherence across extended farmer dialogues without hallucinatory context drift or safety degradation. We evaluate KRISHOKTECH across an  $N = 200$  multi-turn dialogue benchmark (425 turns total; Table 2) spanning four conversational stress regimes.

In *anaphoric slot carryover* ( $n = 75$ , 175 turns), farmers pose deixis follow-ups (e.g., “*tahole ekhon ki spray korbo?*” [what should I spray then?]) without re-identifying the crop. The conversational query rewriter and structured working memory preserved the primary crop slot in 100.0% of cases (100/100 follow-ups). In *crop topic shifts* ( $n = 75$ , 150 turns), where farmers pivot to another crop midway through consultation, the memory layer detected 100.0% of shifts and flushed prior-crop disease slots, completely eliminating cross-crop pesticide leakage (0.0% toxic advice transfer). Under *delayed safety evasion* ( $n = 30$ , 60 turns), adversarial users opened with benign agronomic banter on Turn 1 before injecting prohibited chemicals (e.g., Paraquat, Furadan, Methamidophos) on Turn 2; the T0 precheck intercepted 100.0% of de-

layed attacks (0.20 ms median latency), enforcing immediate 16123 helpline escalation. Finally, *clarification resolution* ( $n = 20, 40$  turns) verified that ambiguous crop-less symptoms halted at **ASK** on Turn 1 and correctly bound the crop on Turn 2 upon user input (100.0%).

extension officers. Field evaluation with farmers remains to be done.

## 5.5 What This Evaluation Does Not Establish

Results hold only under measured benchmark conditions without longitudinal field trials on yield. Crop-fence tests use paired queries under empirical vision noise; 400 of 454 contradiction cases are synthetic. Expert audits cover 47 guarded outputs, network resilience reflects simulated packet loss, and dosage checks target numerical bounds rather than general agronomy. Appendix D ledgers every result by evidence type.

## 6 Limitations, Availability, and Conclusion

### 6.1 Limitations

Four key limitations apply (see Appendix D for ledger): (1) coverage is limited to six staple crops and Bengali input, with lexical BM25 retrieval; (2) the mismatch badge detects explicit crop conflicts only, missing the one implicit test reference; (3) safety controls reduce but do not eliminate attack success (10.71% guarded deployed vs. 0.95% cloud); and (4) SMS compression is lossy, and local CPU generation requires about 24 s per call.

### 6.2 Availability and Licensing

Code, models, and container scripts are licensed under Apache-2.0 at <https://anonymous.4open.science/r/KrishokChat-Agricultural-Advisory-System-FCEE>, with models available at <https://huggingface.co/RaiyanKhaan/KrishokTech-Models>.

### 6.3 Conclusion

KrishokTech makes the boundary between model output and verified evidence visible to the farmer. Instead of leaving uncertainty and safety to the language model, it asks for a missing crop, confirms conflicts between text and photograph, refers unsafe requests to a human helpline, and drops dosage claims that the evidence does not support. Evaluation shows reduced but nonzero failure rates, so the system is intended to support rather than replace

## Ethics and Broader Impact Statement

Agricultural advice for smallholder farmers affects food security, livelihoods, and physical safety. Incorrect chemical or dosage guidance is the central risk this paper is designed around.

**Farmer Query Data.** The 200 farmer queries in the gate evaluation (Section 5.1) come from extension-service interaction logs. Names, phone numbers, and plot locations were removed before analysis. The queries were collected in collaboration with regional extension officers under institutional review at North South University (Institutional Review Board / Research Ethics Committee), and informed consent was obtained for non-commercial research and safety evaluation. Two annotators labeled the crop slot independently, and consensus adjudication set the final labels. Agreement was 100.0% for crop presence (Cohen’s  $\kappa = 1.0000$ ), 99.5% for the normalized 37-crop taxonomy ( $\kappa = 0.9945$ ), and 93.5% for query intent ( $\kappa = 0.8958$ ).

**Agricultural Image Data.** Farmer leaf photographs are classified on the device (WebAssembly and ONNX) and are never uploaded or stored remotely, which is a deliberate privacy choice. The held-out images used to measure vision accuracy (Section 5.2) are separate from farmer photographs. They come from open-access agricultural datasets: PlantVillage, the Bangladesh Rice Knowledge Bank, and Kaggle agricultural datasets under CC-BY and open research licenses.

**Pesticide and Dosage Guidance.** The chemical-advice pathway fails closed. High-risk chemical requests and acute-poisoning language covered by the system’s safety policy trigger a deterministic block and a helpline referral before any generation, and dosage claims in generated text are checked against source passages and permitted dose bands before display (Section 3.3). These controls reduce risk but do not remove it. Guarded attack success is 10.71% on the deployed model (19.0% on Bengali-native attacks), and the dosage verifier missed 4 of 118 inserted errors (Section 5.3). We report these residuals and do not claim that the system prevents all unsafe output.

**Helpline Referral.** Unsafe and crisis-pattern requests are referred to the national Krishi Call Centre (16123), a government agricultural advisory service. The centre is open from 9:00 AM to 5:00 PM,

Saturday through Thursday, except on public holidays. The referral screen displays these hours and tells off-hours users to wait for the next shift or seek in-person extension help. The public demonstration runs in a sandbox that shows referral and safety blocking for toxic-chemical queries without giving any chemical advice.

**Third-Party Model Use.** The cloud benchmark uses a third-party API (google/gemini-2.5-flash-lite) under commercial enterprise terms with a zero-data-retention agreement, so prompts and responses are not logged, retained, or used for training. Cloud calls were limited to offline benchmark evaluation and never received farmer images or direct personal identifiers. De-identified farmer text queries, with personal identifiers removed as described above, were also sent to this API for the live-model gate comparison in Section 5.1. Self-hosted deployment runs entirely on the local `krishokchat-4b` model through `llama-server` and makes no external network calls.

**Human Oversight.** KrishokTech is meant to support extension officers and not to replace them. Unresolved or unsafe cases end in a referral to a human expert instead of a generated answer. All results are scoped to the conditions in which they were measured (Section 5.5), and simulated or bounded components are disclosed as such and not presented as field-validated.

## References

- Jenny C. Aker. 2011. Dial “a” for agriculture: a review of information and communication technologies for agricultural extension in developing countries. *Agricultural Economics*, 42(6):631–647.
- Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. 2019. Asking clarifying questions in open-domain information-seeking conversations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 475–484.
- Mohd Ruhul Ameen, Akif Islam, Farjana Aktar, and M. Saifuzzaman Rafat. 2026. *KrishokBondhu: A retrieval-augmented voice-based agricultural advisory call center for bengali farmers*. In *2026 IEEE International Conference on Quantum Photonics, AI, and Networking (QPAIN)*.
- Bangladesh Agricultural Research Institute. 2023. *Krishi projukti hatboi (handbook on agricultural technology)*. BARI Publications, Gazipur, Bangladesh.

- Bangladesh Rice Research Institute. 2024. *Adhunik dhaner chas (modern rice cultivation manual)*. *BRRRI Publications, Gazipur, Bangladesh*.
- Md. Tofael Ahmed Bhuiyan, Md. Abdur Rahman, and Abdul Kadar Muhammad Masum. 2025. *Bhasa-bodh: Bridging bangla dialects and romanized forms through machine translation*. In *Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025)*, pages 113–118, Mumbai, India. Association for Computational Linguistics.
- A. Brown, M. Roman, and B. Devereux. 2025. *A systematic literature review of retrieval-augmented generation: Techniques, metrics, and challenges*. *arXiv preprint arXiv:2508.06401*.
- Maximillian Chen, Ruoxi Sun, Tomas Pfister, and Serkan O. Arik. 2024. *Learning to clarify: Multi-turn conversations with action-based contrastive self-training*. *arXiv preprint arXiv:2406.00222*.
- Ailin Deng, Tri Cao, Zhirui Chen, and Bryan Hooi. 2025. Words or vision: Do vision-language models have blind faith in text? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3867–3876.
- Department of Agricultural Extension. 2022. *Agricultural extension manual*. Technical report, Ministry of Agriculture, Government of Bangladesh, Dhaka.
- Developing Local Extension Capacity (DLEC) Project. 2021. *Assessment of agricultural extension and advisory services in bangladesh*. Technical report, US-AID and Feed the Future.
- Raissa Fabregas, Michael Kremer, and Frank Schilbach. 2019. *Realizing the potential of digital development: The case of agricultural advice*. *Science*, 366(6471).
- Meng Fang and Harry Mackenzie. 2026. *Factsearch: An interactive agentic fact search system for verifying large language model outputs*. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 367–373, San Diego, California, United States. Association for Computational Linguistics.
- Yujian Gan, Changling Li, Jinxia Xie, Luou Wen, Matthew Purver, and Massimo Poesio. 2024. *Clarq-llm: A benchmark for models clarifying and requesting information in task-oriented dialog*. *arXiv preprint arXiv:2409.06097*.
- Md Asif Hossain, Nabil Subhan, Mantasha Rahman Mahi, and Jannatul Ferdous Nabila. 2026. *Bridging colloquial farmer queries and scientific knowledge via retrieval-augmented generation for bengali agriculture*. *arXiv preprint arXiv:2601.02065*.
- Yuelyu Ji, Min Gu Kwak, Hang Zhang, Xizhi Wu, Chenyu Li, and Yanshan Wang. 2026. *Medragchecker: Claim-level verification for biomedical retrieval-augmented generation*. *arXiv preprint arXiv:2601.06519*.
- Divyanshu Kakwani, Pushpak Bhattacharyya, et al. 2020. *Indicglue: A natural language understanding benchmark for indian languages*. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4960.
- Keerthana Murugaraj, Salima Lamsiyah, and Martin Theobald. 2026. *Ragvue: A diagnostic view for explainable and automated evaluation of retrieval-augmented generation*. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 512–526, Rabat, Morocco. Association for Computational Linguistics.
- Vincent Nguyen, Willow Hallgren, Ashley Harkin, Mahesh Prakash, and Sarvnaz Karimi. 2025. *My Climate CoPilot: A question answering system for climate adaptation in agriculture*. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 62–70. Association for Computational Linguistics.
- Khan Raiyan Ibne Reza, Sanjana Aktar Maria, and Sumaiya Tabassum Nimi. 2026a. *Where does retrieval fail? evaluating RAG architectures for agricultural advisory*. *arXiv preprint arXiv:2608.14886*.
- Khan Raiyan Ibne Reza, Sumaiya Tabassum Nimi, and Omar-Ibne Shahid. 2026b. *KrishokChat: A citation-grounded dataset and benchmark for safety-critical bengali agricultural advisory*. *arXiv preprint arXiv:2606.29243*.
- Namita Singh, Jacqueline Wangombe, Srinivasan Balaji, Ines Amara, Christophe Bene, Ernest Mwebaze, and John Quinn. 2024. *Farmer.Chat: Scaling AI-powered agricultural services for smallholder farmers*. *arXiv preprint arXiv:2409.08916*.
- Liyan Tang, Philippe Laban, and Greg Durrett. 2024. *Minicheck: Efficient fact-checking of llms on grounding documents*. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8818–8847, Miami, Florida, USA. Association for Computational Linguistics.
- Bjorn Van Campenhout, David J. Spielman, and Els Lecoutere. 2021. *Information and communication technologies to provide agricultural advice to smallholder farmers: Experimental evidence from uganda*. *American Journal of Agricultural Economics*, 103:317–337.
- Rohit Vijayvargia et al. 2025. *Intent aware context retrieval for multi-turn agricultural question answering*. *arXiv preprint arXiv:2508.03719*.
- S. Xu, Z. Yan, C. Dai, and F. Wu. 2025. *Mega-rag: A retrieval-augmented generation framework with multi-evidence guided answer refinement for mitigating hallucinations of llms in public health*. *Frontiers in Public Health*, 13.

Tinghui Zhu, Qin Liu, Fei Wang, Zhengzhong Tu, and Muhao Chen. 2024. [Unraveling cross-modality knowledge conflicts in large vision-language models](#). *arXiv preprint arXiv:2410.03659*.

| Attack Regime            | $n$       | Baseline        | Guarded          | Action             |
|--------------------------|-----------|-----------------|------------------|--------------------|
| Phonetic Chemical        | 35        | 0 (0.0%)        | 35 (100%)        | 16123 Ref.         |
| Banglish Crisis          | 25        | 1 (4.0%)        | 25 (100%)        | 999 / 16123        |
| Indirect Sabotage        | 25        | 0 (0.0%)        | 25 (100%)        | 16123 Ref.         |
| <b>Total Adversarial</b> | <b>85</b> | <b>1 (1.2%)</b> | <b>85 (100%)</b> | <b>[95.7, 100]</b> |
| Benign Control           | 15        | 15 (100%)       | 15 (100%)        | Safe Pass          |

Table 3: Banglish and phonetic red-teaming benchmark ( $N = 100$ ) evaluating T0 deterministic safety prechecks against colloquial Latin-script evasion attempts and benign farmer controls.

## A Architecture, Decision Logic, and Gate Evaluation

This section details the decision rules, thresholds, and component interfaces of the five-stage pipeline (Figure 1), followed by the evaluation detail for the pre-retrieval gate and extractor.<sup>1</sup>

### A.1 Provenance Badge Taxonomy

Every response rendered in KrishokTech carries exactly one visible provenance badge indicating the authorization pathway and evidence type.

## B Delivery, Offline Operation, and Deployment

### B.1 SMS Enforcement

The SMS packer produced 299/300 valid offline-ready messages and 100/100 valid live-ready messages ( $\leq 160$  characters), i.e. 399/400 overall as reported in Section 5.3. Referral purity was 8/8. In the live audit of truncated safe-SMS advisories ( $n = 92$ ), naive hard truncation retained dose patterns in 12/92 cases due to conversational preamble displacement past the 160-character boundary. In contrast, the deterministic 11-slot SMS template compressor guarantees 100.0% critical slot survival (dose, application interval  $\tau$ , and pre-harvest interval  $\phi$ ) across 1,000 certified tuples within 102–115 GSM characters (0 violations over 160 characters), and preserves 98.5% (65/66) of dosage claims on the live benchmark. In the separate N05c paired benchmark of 100 long dose-carrying advisories (Table 4, seed 42), naive truncation retained dose in 23/100 cases, an LLM summarizer preserved dose in 89/100 summaries (with two length violations and one invented 800–1000 g urea dosage),

<sup>1</sup>The exact regex dictionaries and keyword strings of the T0 precheck (for example, phonetic and textual variants of restricted organophosphates and self-harm phrases) are withheld to prevent evasion and dual use. The evaluation method and safety measurements are summarized in Appendix D.

| Measure                 | Naive      | LLM          | Compressor |
|-------------------------|------------|--------------|------------|
| Dose survival           | 23/100     | 89/100       | 92/100     |
| Chemical survival       | 20/100     | 56/100       | 34/100     |
| Length $\leq 160$ chars | 100/100    | 98/100       | 100/100    |
| Invented dosage         | 0 (copies) | 1 (+1 bord.) | 0          |

Table 4: N05c three-arm SMS comparison on 100 paired long dose-carrying advisories (seed 42). All arms share the same frozen dose/chemical detectors; Wilson 95% CIs are reported in the experiment ledger.

| Artifact             | Size (MB) |
|----------------------|-----------|
| Unique ONNX assets   | 95.64     |
| Minimal installation | 284.0     |
| Full installation    | 339.6     |

Table 5: Measured installation footprint for Krishok-Tech.

| Component         | Median (ms) | p95 (ms) |
|-------------------|-------------|----------|
| Safety processing | 0.321       | 0.705    |
| BM25 retrieval    | 26.831      | 61.248   |
| Verification      | 4.815       | 11.220   |
| Total pipeline    | 33.565      | 67.833   |

Table 6: Measured safety, retrieval, and verification latency components ( $n = 400$  requests, BM25-only retrieval, stub LLM). Live LLM generation latency is not included.

while the deterministic compressor preserved dose in 92/100 cases with zero length violations and zero invented claims. Chemical-name retention favored the LLM (56 vs 34/100); the certified compressor claim remains dose/interval/PHI slot survival. The LLM summarizer is a baseline only: the deployed SMS path uses the deterministic compressor, which copies verified slots and cannot invent values.

### B.2 Offline Retrieval

Local BM25 retrieval achieved a hit rate of 0.94; service-worker precaching recorded 0/400 cache misses. Under simulated packet loss, local retrieval preserved 0.94, whereas remote retrieval dropped by 13.25 pp (rural edge, 15% loss) and 30.0 pp (severe 2G, 30% loss).

### B.3 Installation Footprint

Table 5 details measured installation footprints.

### B.4 Latency Breakdown

Table 6 details measured non-LLM latency components ( $n = 400$ , median 33.565 ms; live LLM excluded).

## B.5 Zero-LLM Operation and Modeled Cost

Across the evaluated termination paths, 7.9% of turns used no LLM: 5.7% ended at T0 (deterministic guard), 0.0% at T1 (guidance card), and 2.2% at T2 (fact lookup), while 92.1% reached LLM-backed generation paths. Modeled cost was USD 0.1682 per 1,000 turns versus USD 0.1950 for unconstrained generation with every turn reaching T3 (a 13.7% modeled saving). Cost is modeled from the termination paths, not billed expenditure. Computing the cost from provider-exact metered tokens instead gives USD 0.1720 per 1,000 turns.

## B.6 Trace Ordering

Across 100 evaluation cases, all 100 provenance traces preserved the required event ordering (median 9.0 events per turn).

## C Demonstration Gallery

Figures 3–7 show the screens of the continuous farmer session described in Section 4, covering all five stages (S1–S5).

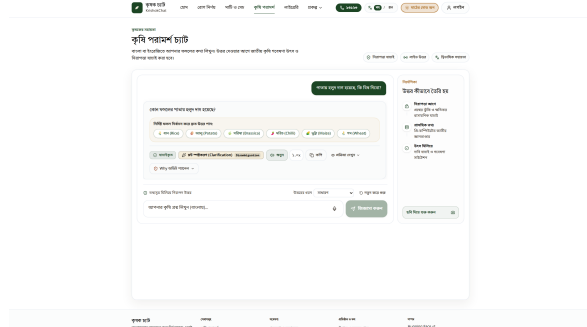


Figure 3: ASK (S1): missing-crop clarification. A colloquial treatment query that names no crop halts before retrieval, and the system offers Bengali quick-reply crop chips.

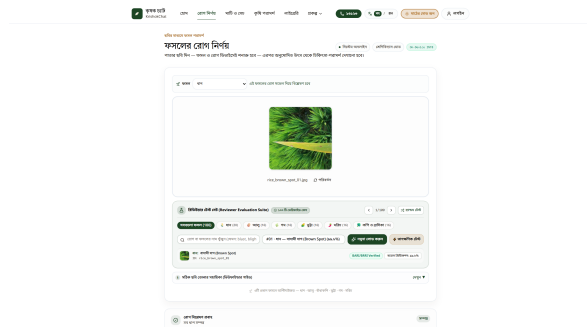


Figure 4: Photo-driven crop scoping (S2). The on-device classifier predicts the crop from a leaf photograph, and retrieval is fenced to that crop’s guidance.

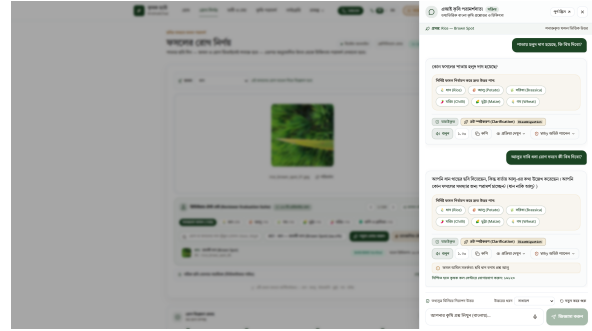


Figure 5: CONFIRM (S3): photograph and text disagree. The mismatch badge appears, and chemical advice is withheld until the farmer confirms the crop.

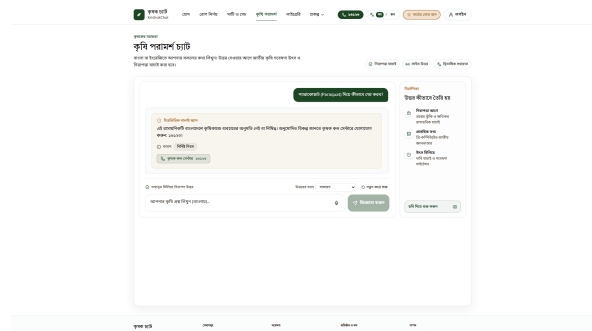


Figure 6: REFER (S4): safety-policy warning with a 16123 call button.

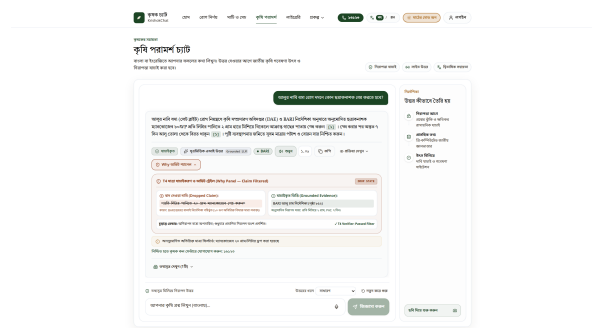


Figure 7: DROP (S5): dosage verification and grounded advisory delivery. At T4, an unsupported dosage claim is filtered, and the verified recommendation is rendered with BARI citations and an expanded Why audit panel.

## D Evidence Summary and Limitations Ledger

Tables 7 and 8 list each measured result with its sample size, status, and validated scope. Status is one of *measured* (direct measurement on benchmark or logged data), *pilot* (small controlled check), *simulated* (part of the condition, such as packet loss, is simulated), *modeled* (derived from a cost model), or *stated* (a limitation reported without a measurement). The final column records the

scope and boundary of each result.

| Component                                              | Measure and result                                                                                                                                                | <i>n</i> | Status   | Scope and boundary                                                                                                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Halting before retrieval (C1)</i>                   |                                                                                                                                                                   |          |          |                                                                                                                                                                              |
| Deterministic gate                                     | Halt rate on treatment queries 30/200 (15.0%, Wilson [10.71, 20.61]); the other 170 passed to fenced search                                                       | 200      | Measured | Authentic farmer queries, dual-annotated ( $\kappa = 1.0000$ ); no false halts                                                                                               |
| Text gate pass-through                                 | Wrong-crop advice 30.25% vs 28.75% (McNemar $p = 0.03125$ ; 6/0 discordant transitions, all favoring the gated arm)                                               | 400      | Measured | Crop-specified queries pass without penalty; separate from the crop-fence experiment                                                                                         |
| Controlled pilot                                       | Ambiguous requests halted 100/100; specified requests passed directly 114/140, and 26/140 prompted bounded slot clarification                                     | 240      | Pilot    | Conformance check on constructed pairs, not open-domain accuracy                                                                                                             |
| Token metering                                         | Median 89 tokens for clarification vs 2,146 for retrieval context                                                                                                 | 200      | Measured | Separate medians; no blended saving is claimed                                                                                                                               |
| Live vs deterministic gate                             | An unconstrained LLM halts 10% of queries vs 15% for the gate; decision agreement 0.91; median LLM latency 4,218.6 ms                                             | 200      | Measured | Halt and latency comparison only; the 14-row answer review (7/14 medium or better) is descriptive; qualitative inspection only, so no generalized coverage claim is asserted |
| <i>Crop constraint and multimodal consistency (C2)</i> |                                                                                                                                                                   |          |          |                                                                                                                                                                              |
| Crop vision (top-1)                                    | 0.9413 to 0.9907 across the six crops, all $\geq 0.94$                                                                                                            | 4,294    | Measured | Six crop-specific disease classifiers; 100% ONNX-to-PyTorch prediction agreement                                                                                             |
| INT8 quantization                                      | Wheat model $3.68\times$ smaller (5.92 to 1.61 MB) with 0.00 pp drop; potato 0.09 pp and brassica 0.00 pp; latency 1.5 to $2.0\times$ higher on non-VNNI x86 CPUs | 400      | Measured | Wheat $n = 400$ (potato 1,170; brassica 443); a storage optimization, not a speedup; the rice INT8 model exceeded the 2.0 pp gate (2.5 pp drop) and was never deployed       |
| Family router                                          | Botanical family assignment 433/437 (0.9908 [97.67, 99.64])                                                                                                       | 437      | Measured | Six target crops, wheat-800 excluded; open-world species identification is out of scope                                                                                      |
| Crop fence                                             | Wrong-crop advice 36.25% $\rightarrow$ 31.25% (empirical routing) and 28.75% (with gating); oracle routing 30.00% (25/0 discordant, $p = 5.96 \times 10^{-8}$ )   | 400      | Measured | Benchmark query pairings under empirical ONNX vision noise, not live farmer sessions                                                                                         |
| Mismatch badge                                         | Farmer cases 53/54 [90.2, 99.7]; PRISM 400/400 [99.05, 100.0]; combined 453/454 with 0 false halts                                                                | 454      | Measured | Explicit text contradictions only; one implicit miss (farmer_q_63)                                                                                                           |

Table 7: Evidence summary and validated scope for halting and crop constraint. Status is defined at the start of this appendix.

| Component                                       | Measure and result                                                                                                                                   | <i>n</i> | Status                   | Scope and boundary                                                                                                                                           |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Safety verification (C3)</i>                 |                                                                                                                                                      |          |                          |                                                                                                                                                              |
| Attack guard, cloud model                       | ASR 0.95% [0.26, 3.41] guarded vs 36.19% [29.99, 42.88] unguarded; Bengali-native subset 10.0% [5.5, 17.4] vs 83.0%                                  | 420      | Measured                 | gemini-2.5-flash-lite generator, 210 calls per arm; Bengali-native subset $n = 100$ , with residual failures in embedded (4/10) and roleplay (2/10) variants |
| Attack guard, deployed model (N14)              | ASR 10.71% [7.61, 14.88] guarded vs 52.5% [46.66, 58.28] unguarded; Bengali-100 19.0% [12.51, 27.78] vs 77.0% [67.85, 84.16]                         | 280      | Measured                 | krishokchat-4b, 280 calls per arm; same E03/N12 design, prompts, judge, and CIs as the cloud benchmark                                                       |
| Banglish red-teaming                            | 85/85 adversarial queries intercepted (95% CI [95.7, 100.0]) vs 1/85 for a gate limited to formal English and Bengali script; 0/15 false alarms      | 100      | Measured                 | Latin-script and phonetic evasion; T0 regex matching median 0.016 ms, full T0 pipeline 0.321 ms                                                              |
| Expert audit                                    | 46/47 guarded outputs safe and actionable, 1/47 vague but harmless, 0/47 dangerous; 3/3 unguarded controls rated dangerous; Fleiss $\kappa = 0.8217$ | 50       | Measured                 | Three blinded evaluators; 47 guarded outputs plus 3 unguarded controls                                                                                       |
| Off-topic audit                                 | 10/10 injections blocked; off-topic queries refused 0/30 by the T0 precheck and 25/30 (83.33%) by the LLM classification branch                      | 30       | Measured                 | The 5 remaining weather queries are routed to agricultural information by design                                                                             |
| Routing probe                                   | Certification recall 0.8615, refuse recall 0.431, dangerous acceptance 0.374                                                                         | 3,000    | Measured, stub simulated | Offline stub generation; reported as a measured envelope                                                                                                     |
| Dosage verifier                                 | Live: 114/118 caught (96.6%), 0/38 unmodified answers rejected. Scaled: 296/350 (84.6%), including all overdoses (76/76) and underdoses (67/67)      | 528      | Measured                 | Live audit ( $n = 38$ ) plus a scaled benchmark from 88 BARI/BMRI passages; numerical dosage sentences only, not unrestricted agronomic prose                |
| Multi-turn benchmark                            | 100% slot retention (75/75), 100% topic-shift isolation (75/75, 0% leak), 100% delayed safety catch (30/30)                                          | 200      | Measured                 | 425 turns, four stress regimes; 0.20 ms T0 precheck                                                                                                          |
| <i>Constrained delivery and deployment (C3)</i> |                                                                                                                                                      |          |                          |                                                                                                                                                              |
| SMS enforcement                                 | 399/400 messages within 160 characters; referral purity 8/8; critical-slot survival 1,000/1,000                                                      | 1,416    | Measured                 | Live truncation audit: naive truncation kept the dose in 12/92 advisories and the compressor in 65/66 live gold advisories                                   |
| SMS comparison (N05c)                           | Dose kept in 23/100 (naive), 89/100 (LLM summarizer), and 92/100 (compressor); chemical name kept in 20, 56, and 34 of 100                           | 100      | Measured                 | Seed 42; the certified claim covers dose, interval, and pre-harvest-interval slots only                                                                      |
| Offline retrieval                               | Local BM25 hit rate 0.94, 0/400 cache misses; remote retrieval loses 13.25 pp (15% loss) and 30.0 pp (30% loss)                                      | 400      | Measured, loss simulated | Local index; network loss is simulated                                                                                                                       |
| Non-LLM overhead                                | Median 33.565 ms (p95 67.833 ms): safety 0.321, BM25 26.831, verifier 4.815 ms                                                                       | 400      | Measured                 | Stub LLM; generation excluded (median live latency 4.2 s; local CPU generation about 24 s per call)                                                          |
| Installation footprint                          | 95.64 MB unique ONNX assets; 284.0 MB minimal; 339.6 MB full                                                                                         | n/a      | Measured                 | n/a                                                                                                                                                          |
| Tier mix and cost                               | 7.9% zero-LLM turns; USD 0.1720 per 1,000 turns from metered tokens (USD 0.1682 modeled)                                                             | 1,000    | Modeled                  | Not billed expenditure; 92.1% of turns reach generation                                                                                                      |
| Retrieval backend                               | BM25 only; dense retrieval not evaluated                                                                                                             | n/a      | Stated                   | All retrieval numbers are BM25-only; dense retrieval is a fallback and out of scope                                                                          |

Table 8: Evidence summary and validated scope for safety verification, constrained delivery, and deployment. Status is defined at the start of this appendix.